

Automation of Weather-Based Sales and Staffing Decisions

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Problem

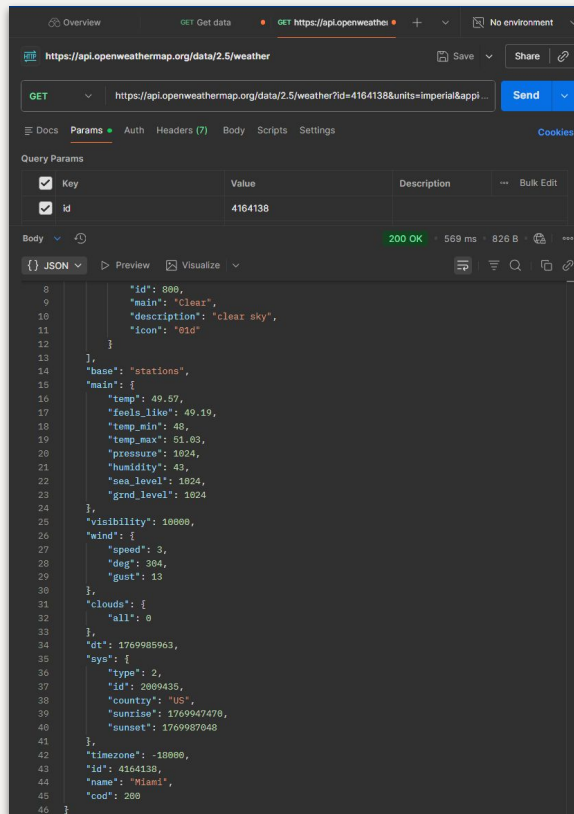
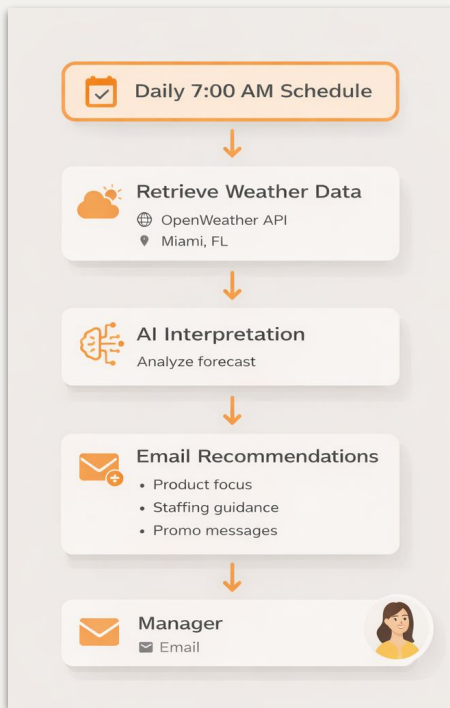
Bayside Brews & Scoops operates in a highly weather-dependent environment, where daily temperature and conditions directly influence customer traffic, product demand, and staffing needs. Currently, weather checks and resulting decisions are handled manually, often just before opening, leading to inconsistent preparation and delayed adjustments. This makes it difficult to reliably align inventory, promotions, and staffing with expected demand on a day-to-day basis.

Solution

An automated workflow retrieves daily weather data each morning and generates actionable recommendations before the business opens. Using real-time forecast information, the system produces clear guidance on product focus, staffing needs, and promotional messaging, then delivers this summary directly by email. This ensures decisions are timely, consistent, and aligned with expected demand without requiring manual checks.

Workflow diagram

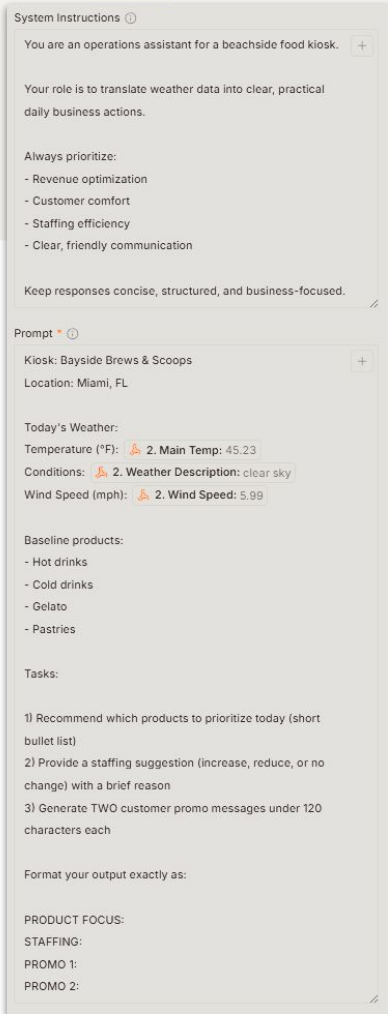
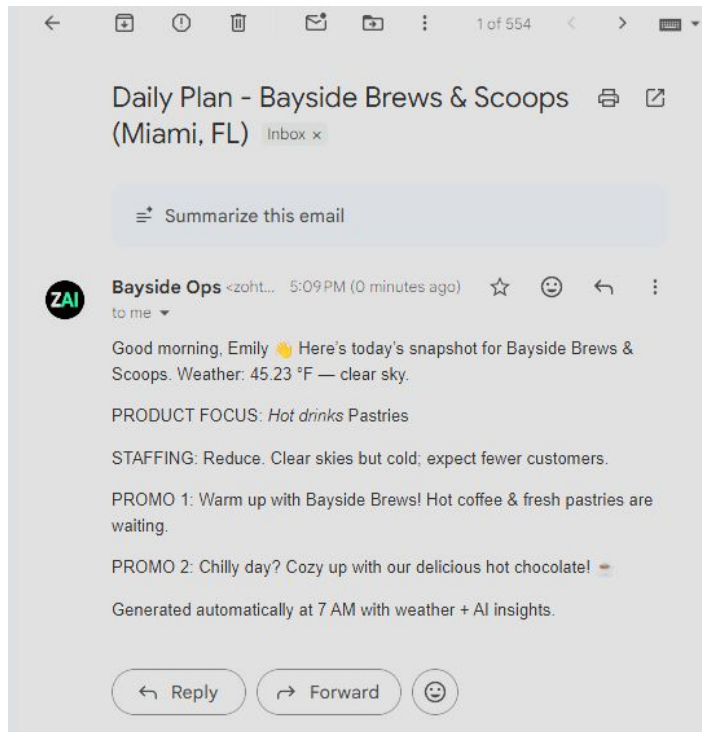
This automation runs on a daily schedule and begins by retrieving current weather data through a custom API request, which was validated in Postman prior to implementation. The workflow then evaluates the forecast and routes the data through predefined paths to generate the appropriate output. Using AI, the system interprets weather conditions to produce business-ready recommendations, which are formatted into a clear message and delivered automatically by email. This ensures that each morning's decisions are based on consistent inputs and follow a repeatable process without requiring manual intervention.



Workflow in action

The screenshot on the right shows the AI prompt and structured input data used by the automation. The email on the left shows the final output delivered to the manager.

Together, these demonstrate the full workflow from weather data intake to decision-ready communication, showing how forecast information is translated into clear product, staffing, and promotional guidance without manual intervention.



Benefits

Operational Efficiency

- Eliminates manual weather checks
- Automates daily decision prep before opening
- Reduces last-minute adjustments and guesswork

Consistent Decision-Making

- Uses the same data inputs every day
- Applies repeatable logic to staffing and product focus
- Reduces variability between shifts or managers

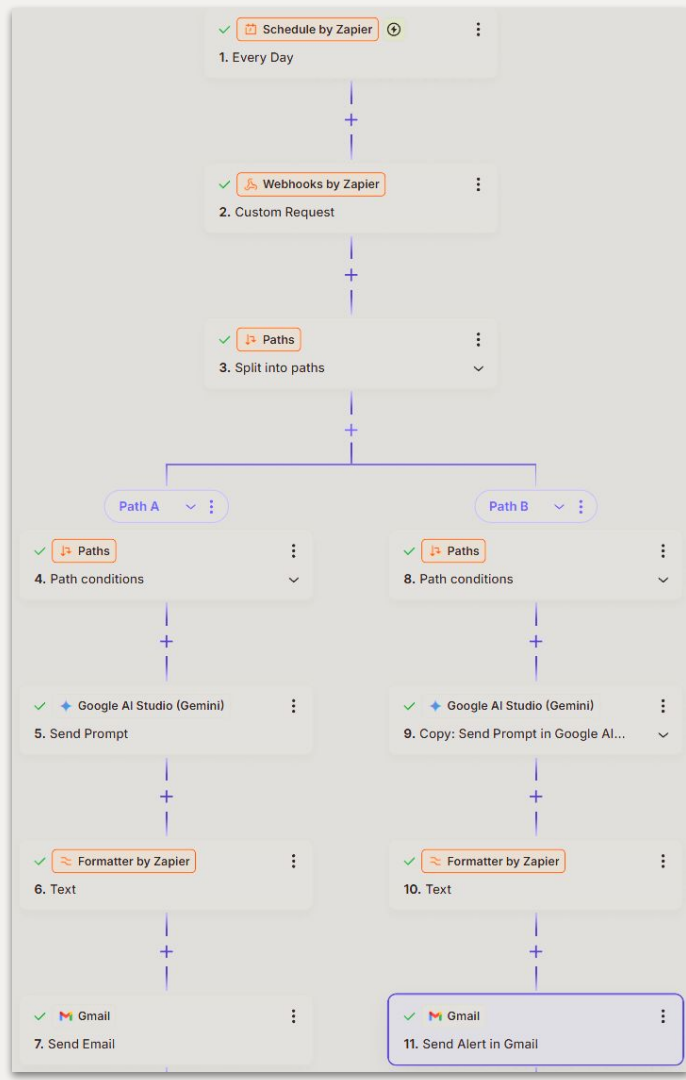
Actionable Communication

- Delivers clear, structured guidance by email
- Translates forecasts into practical recommendations
- Supports faster, more confident day-of decisions

Optional extras

Severe Weather Alert (Applied)

A severe weather alert is already implemented within the Zapier workflow using conditional paths. When temperatures exceed **91°F** or when weather conditions include **rain** or **storms**, the workflow routes to a dedicated alert email. This ensures timely operational adjustments during rapidly changing coastal weather conditions, supporting staff readiness, customer comfort, and service quality.



Challenges & improvements

A key challenge in designing this workflow was working with limited insight into the kiosk's real-world operations and strategic goals. Assumptions were required around the product mix, preparation time, inventory constraints, staffing coordination, and how daily decisions are communicated to the team. In addition, the absence of clear business objectives—such as whether the priority is revenue growth, cost control, customer experience, or expansion—limited how far recommendations could be optimized or scaled. To keep the system reliable and within scope, the workflow was intentionally focused on weather-driven decision support rather than deeper execution or optimization logic. With additional time, tools, or access to business data, the automation could be improved by aligning recommendations more closely with defined business goals, validating product and inventory details, integrating staff communication channels, and extending the workflow to support multiple kiosks or locations. These enhancements would allow the system to scale more effectively while preserving its core, repeatable automation design.